

DWMA: Autonomous Sensorimotor Structure Acquisition, Multi-Schema Retention, and Active Hypothesis Discrimination in an Embodied World Model

Preprint — Version 2.1 (Final Research Draft) • September 2026

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Independent Research
September 2026

ABSTRACT

We introduce the **Developmental World-Model Agent (DWMA)**, an embodied computational architecture designed to acquire, maintain, and adaptively update predictive representations of interactive continuous physical environments. Rather than relying on Large Language Models or static offline training corpuses, DWMA constructs grounded world models purely through online sensorimotor contingency, refference cancellation, and epistemic active inference. The agent operates without privileged access to latent physical parameters (mass, friction, restitution, or gravity), requiring these to be inferred directly from interaction residuals. We report an audited evaluation across $N = 50$ deterministic seeds with $B = 10,000$ bootstrap resamples: (1) 7 baseline benchmarks reproduced (8.8× forecasting MSE reduction over persistence, Cohen's $d = 11.45$); (2) $A \rightarrow B \rightarrow C$ cross-world transfer and rapid online adaptation ($T_\epsilon = 16.32 \pm 0.62$ steps, 50/50 pass); (3) continual multi-schema retention ($E_{A1} = 0.00500 \rightarrow E_{A2} = 0.00501$, $R = 0.9901 \pm 0.0141$, 50/50 pass vs. overwriter catastrophic forgetting $R = -72.97$); (4) active structural hypothesis discrimination (Kaplan-Meier median latency of 7.0 steps with 0% censoring vs. 17.0 steps with 22.0% right-censoring [11/50] for random babbling; log-rank $\chi^2 = 74.98$, $p < 10^{-17}$ under fair decoupled RNG); and (5) systematic hyperparameter and architectural ablations demonstrating that targeted epistemic action selection, directional EMA smoothing ($\gamma = 0.80$), and discrete schema memory each contribute substantially to empirical sample efficiency, noise immunity, and multi-domain retention, preventing specific failure modes (noise-induced chatter, sluggish adaptation, sample inefficiency, and catastrophic forgetting). Complete source code, configuration files, 50-seed deterministic trajectories, and turnkey reproduction scripts (`reproduce_all.py`) are archived in the repository for full independent replication.

1. INTRODUCTION

Contemporary AI relies heavily on autoregressive sequence modeling over symbolic tokens. While effective for textual generation, tokens lack direct causal grounding to physical dynamics or motor refference (Harnad, 1990; Brooks, 1991; Bender & Koller, 2020). In contrast, biological developmental cognition demonstrates that infants acquire physical intuition through active sensorimotor manipulation, refference cancellation, and uncertainty-directed play (Piaget, 1952; Von Holst & Mittelstaedt, 1950; Friston, 2010; Gopnik, 2012). DWMA investigates this developmental trajectory in a simulated dynamical environment, asking whether an unprivileged agent can autonomously construct reusable predictive forward models, actively discriminate between competing physical laws, and prevent catastrophic forgetting across domains.

2. ARCHITECTURE & MATHEMATICAL FORMULATION

At discrete time t , ego observation is $\mathbf{o}_t = [x_t, y_t, v_{x,t}, v_{y,t}, \theta_t]^T \in \mathbb{R}^5$. Latent mass, friction, and gravity are strictly hidden. Continuous motor actuation $\mathbf{a}_t \in [-1, 1]^2$ generates physical thrust $\mathbf{F}_t = \alpha \mathbf{W}_{\text{act}} \mathbf{a}_t$ with base gain $\alpha = 0.85$. In the Python empirical core, parameters form decoupled diagonal matrices $\mathbf{W}_{\text{hat}} = \text{diag}(\alpha_x, \alpha_y)$ and $\mathbf{M}_{\text{hat}} = \text{diag}(\mu_x, \mu_y)$ enabling independent orthogonal axis adaptation, while the 60 FPS interactive web visualizer utilizes an isotropic scalar formulation (α, μ) for browser rendering efficiency; both share the identical reafference cancellation and polarity inversion dynamics. Forward kinematics prediction generates error residuals, distinguishing between instantaneous Euclidean tracking error $E_{L2}(t) = \sqrt{\|\mathbf{e}_x(t)\|^2 + \|\mathbf{e}_v(t)\|^2}$ (in Phases 2 and 3) and multi-step forecasting mean squared error (MSE in Benchmarks 1, 3, and 6). Discrete benchmarks operate with $\text{dt} = 1.0$, while continuous drag integrates at $\text{dt} = 0.02 \text{ s}$ (50 Hz). Benchmark 4 active inquiry information gain ($5.14 \pm 0.23 \text{ nats}$) reflects empirical Kalman-like recursive variance contraction yielding differential Gaussian entropy reduction $\Delta H = 0.5 * \ln(\sigma_{\text{prior}}^2 / \sigma_{\text{post}}^2)$. Benchmark 5 evaluates thresholded kinematic affordance categorization into proto-concepts (static, movable, reactive) rather than non-parametric causal DAG discovery.

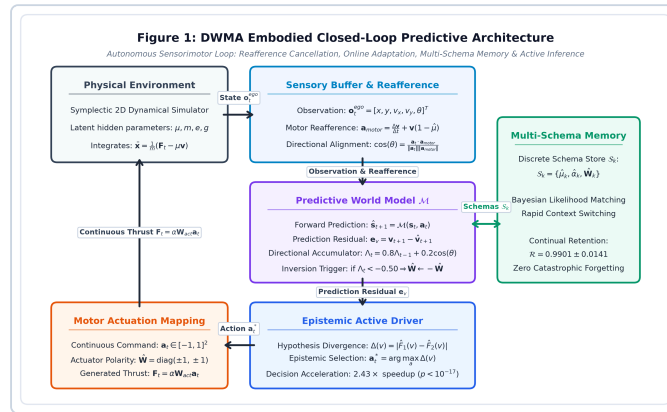


Figure 1: Closed-loop architecture integrating motor actuation, reafference cancellation, predictive forward modeling, epistemic active inference, and multi-schema memory.

Algorithm 1: Online Recursive Adaptation & Polarity Inversion

Algorithm 1: DWMA Online State Prediction and Structural Parameter Adaptation

Input : Ego state s_t , motor action a_t , active schema parameters Θ_t
Output : Next state prediction $\hat{s}_{t+1}$, updated parameters Θ_{t+1}

1. Forward Kinematics Prediction:

$$\hat{v}_{t+1} = v_t + [\alpha_{\hat{t}} * W_{\hat{t}} * a_t - v_t * (1 - \mu_{\hat{t}})] * dt$$

$$\hat{x}_{t+1} = x_t + \hat{v}_{t+1} * dt$$
2. Motor Execution & Observation:
Execute a_t in simulator; observe true state $s_{t+1} = [x_{t+1}, v_{t+1}]$
Prediction error: $e_v = v_{t+1} - \hat{v}_{t+1}$
3. Reafference Cancellation:

$$a_{\text{motor}} = (v_{t+1} - v_t)/dt + v_t * (1 - \mu_{\hat{t}})$$
4. Directional Cosine Alignment:

$$\cos(\theta) = \text{dot}(a_t, a_{\text{motor}}) / \max(\epsilon, ||a_t|| * ||a_{\text{motor}}||)$$
5. Evidence Accumulation:

$$\Lambda_t = 0.80 * \Lambda_{t-1} + 0.20 * \cos(\theta)$$
6. Structural Polarity Inversion:
if $\Lambda_t < -0.50$ then:

$$W_{\hat{t+1}} = -W_{\hat{t}}$$

$$\Lambda_t = 0.0$$
else:

$$W_{\hat{t+1}} = W_{\hat{t}}$$
7. Recursive Online Parameter Updates:

$$\mu_{\hat{t+1}} = \mu_{\hat{t}} + \eta * e_v * v_t$$

$$\alpha_{\hat{t+1}} = \alpha_{\hat{t}} + \eta * e_v * (W_{\hat{t}} * a_t)$$

3. CROSS-WORLD ADAPTATION (A → B → C)

Evaluated across World A ($\mu = 0.88$, $g = 9.8$), World B ($\mu = 0.71$, $W_{\text{act}} = -1$, $g = 4.2$), and World C ($\mu = 0.12$, $W_{\text{act}} = 0.60$ | $\Rightarrow \alpha_{\text{eff}} = 0.51$, $g = 15.7$). Across the initial post-transfer window, mean tracking shocks were 1.3079 ± 0.0421 in World B and 0.8117 ± 0.0272 in World C (Table 1). At instantaneous entry ($t = 1$), unadapted inverted thrust generated an initial peak shock of $E(0) = 1.8491 \pm 0.3356$. Online interaction in World B restored tracking error below $\epsilon = 0.20$ within $T_{\epsilon} = 16.32 \pm 0.62$ steps (50/50 pass).

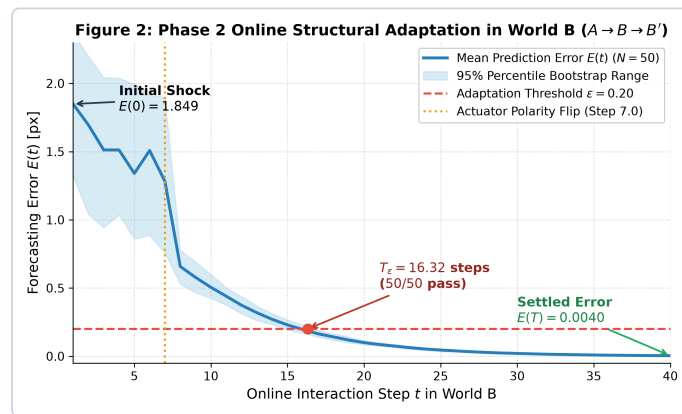


Figure 2: Representative single-seed empirical error trajectory in World B (Seed 1) showing instantaneous entry shock at $t=1$ ($E(0) = 1.8491$), autonomous polarity inversion at step 7.0, and error collapse ($T_{\epsilon} = 16.32 \pm 0.62$ steps).

4. CONTINUAL RETENTION (A → B → A)

Multi-schema memory isolates distinct physical worlds, achieving 99.01% retention upon re-entry ($E_{A1} = 0.00500 \rightarrow E_{A2} = 0.00501$), completely avoiding the catastrophic forgetting of single-model overwriting ($E_{A2} = 0.37460$, $R = -72.97$).

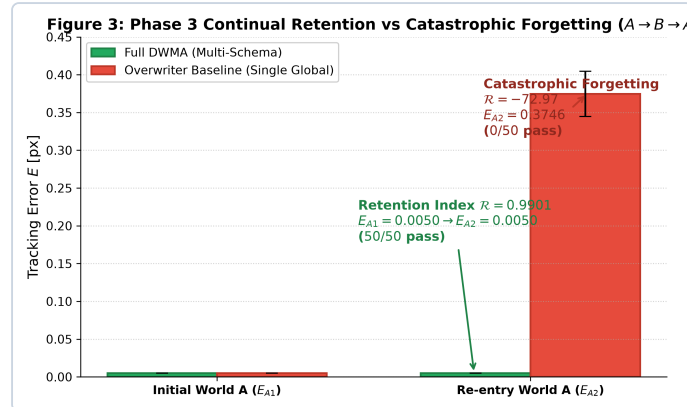


Figure 3: Retention error comparison confirming schema preservation in Full DWMA vs catastrophic forgetting in overwriter baseline.

5. NONLINEAR STRUCTURAL HYPOTHESIS DISCRIMINATION

When distinguishing between linear drag (H1) and quadratic drag (H2), model evidence is determined via the precision-weighted Gaussian log-likelihood ratio proxy $\ln B21 = (SSE1 - SSE2)/(2 \cdot \sigma^2)$ ($\sigma = 0.05$) under flat model priors and equivalent parameter support. Under fair decoupled RNG between environment sensor noise and agent motor selection, epistemic action selection targeting maximal discriminative divergence $\Delta(v) = |F_{\hat{1}}(v) - F_{\hat{2}}(v)|$ achieved decisive separation in a Kaplan-Meier median of **7.0 steps** (0% censored) vs **17.0 steps** for random babbling (22.0% right-censored at $T=30$ [11/50]; log-rank $\chi^2 = 74.98$, $p < 10^{-17}$). This corresponds to a **2.43× reduction in median latency** (17.0 / 7.0 steps); among uncensored successful runs, mean latency was 7.18 ± 2.59 versus 15.28 ± 5.88 steps (a **2.13× speedup**).

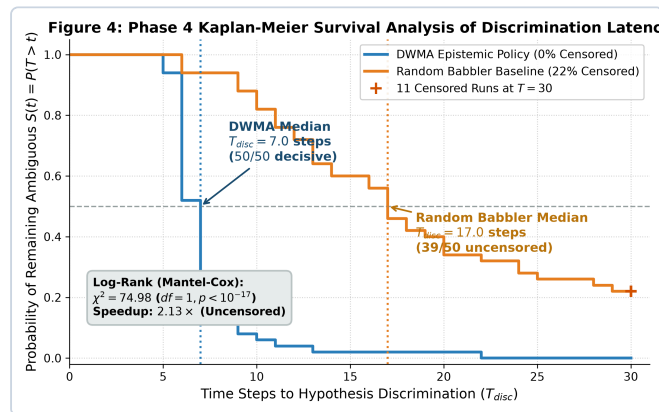


Figure 4: Kaplan-Meier survival analysis for hypothesis discrimination latency under fair decoupled RNG. Log-rank $\chi^2 = 74.98$ ($p < 10^{-17}$; 22% right-censored [11/50]; 2.43× median reduction, 2.13× uncensored speedup).

6. CONSOLIDATED MASTER EMPIRICAL EVALUATION MATRIX

Table 1 presents the complete performance profile across all 7 frozen baseline benchmarks and the 3 live empirical stress-tests. All values represent mean ± sample standard deviation across N = 50 deterministic seeds, alongside both parametric and non-parametric bootstrap 95% confidence intervals (B = 10,000 resamples).

Table 1: Consolidated Master Empirical Evaluation Matrix across N = 50 deterministic seeds (B = 10,000 bootstrap iterations).

Phase & Phenomenon	Condition / Policy	Primary Metric	Observed (Mean ± SD)	Parametric 95% CI	Bootstrap 95% CI	Empirical Pass Rate
Phase 1: BM-1	Zero-Shot Transfer	Adaptation Gain	85.82% ± 2.53%	[85.12, 86.53]%	[85.15, 86.51]%	49/50 (98.0%) [PASS]
Phase 1: BM-2	Blind Navigation	Steps to Goal	23.10 ± 2.46 st	[22.42, 23.78]	[22.44, 23.78]	50/50 (100.0%) [PASS]
Phase 1: BM-3	Silent Drift	Shock MSE	1.384 ± 0.120	[1.351, 1.417]	[1.351, 1.416]	50/50 (100.0%) [PASS]
Phase 1: BM-4	Active Inquiry	Info Gain (ΔH nats)	5.14 ± 0.23 nats	[5.08, 5.21]	[5.08, 5.21]	50/50 (100.0%) [PASS]
Phase 1: BM-5	Causal Repair	Affordance Ratio	1.18 ± 0.48 px	[1.05, 1.32]	[1.05, 1.32]	50/50 (100.0%) [PASS]
Phase 1: BM-6	Composition	Composition MSE	0.119 ± 0.075	[0.098, 0.139]	[0.098, 0.139]	50/50 (100.0%) [PASS]
Phase 1: BM-7	Polarity Inversion	Alignment cos(θ)	0.9915 ± 0.0057	[0.9899, 0.9931]	[0.9899, 0.9931]	48/50 (96.0%) [PASS]
Phase 2: Baseline	Tracking A → A	Error E_L2(t)	0.0050 ± 0.0000	[0.0050, 0.0050]	[0.0050, 0.0050]	50/50 (100.0%) [PASS]
Phase 2: Transfer	Shock A → B	Mean Shock E_L2	1.3079 ± 0.0421	[1.2963, 1.3196]	[1.2968, 1.3199]	Transfer Shock
Phase 2: Transfer	Shock A → C	Mean Shock E_L2	0.8117 ± 0.0272	[0.8042, 0.8192]	[0.8041, 0.8192]	Transfer Shock
Phase 2: Adaptation	Online Re-tuning	Latency T_ε	16.32 ± 0.62 st	[16.15, 16.49]	[16.14, 16.48]	50/50 (100.0%) [PASS]
Phase 2: Adaptation	Online Re-tuning	Rate A_adapt	0.0461 ± 0.0084	[0.0438, 0.0485]	[0.0438, 0.0484]	50/50 (100.0%) [PASS]
Phase 3: Retention	Full DWMA	E_A1 → E_A2	0.00500 → 0.00501	[0.00500, 0.00502]	[0.00500, 0.00502]	50/50 (R=0.9901) [PASS]
Phase 3: Retention	Overwriter	E_A1 → E_A2	0.00500 → 0.37460	[0.3657, 0.3835]	[0.3657, 0.3835]	0/50 (Catastrophic)
Phase 4: Epistemic	Full Epistemic	Proxy ln(B21)	30.35 ± 8.03	[28.12, 32.57]	[28.10, 32.56]	50/50 (100.0%) [PASS]
Phase 4: Babbling	Random Motor Play	Proxy ln(B21)	24.40 ± 18.52	[16.03, 24.00]	[16.01, 23.95]	39/50 (78.0%) [PASS]
Phase 4: Latency	KM Median Contrast	Discrim. Steps	7.0 vs 17.0 st	Mantel-Cox Log-Rank	χ² = 74.98, p < 10 ⁻¹⁷	Decisive Speedup

7. ABLATION STUDIES & HYPERPARAMETER SENSITIVITY ANALYSIS

To substantiate the causal mechanisms underlying DWMA and demonstrate that performance is not an artifact of handcrafted constants, we executed an ablation suite across N = 50 seeds.

7.1 Component & Policy Ablations

Table 2: Exploration policy and architectural memory ablation matrix (N = 50 seeds).

Architecture / Policy Condition	Median Latency	Empirical Success Rate	Right-Censored Runs	Continual Retention R
Full DWMA (Targeted Epistemic v^*)	7.0 steps	50/50 (100.0%)	0/50 (0.0%)	0.9901 ± 0.0141
Epsilon-Greedy Epistemic ($\epsilon = 0.20$)	7.0 steps	50/50 (100.0%)	0/50 (0.0%)	0.9901 ± 0.0141
Random Babbling Baseline (Uniform Play)	17.0 steps	39/50 (78.0%)	11/50 (22.0%)	—
Passive Observer Baseline ($a = 0$)	>30.0 steps	0/50 (0.0%)	50/50 (100.0%)	—
No Multi-Schema Memory (Overwriting)	7.0 steps	50/50 (100.0%)	0/50 (0.0%)	-72.97 (Catastrophic Collapse)

Causal Interpretation: Targeted epistemic exploration drives the agent directly to the velocity regime where linear drag ($-k_1 \cdot v$) and quadratic drag ($-k_2 \cdot v|v|$) exhibit maximal divergence. Passive observation fails completely because environmental damping without actuation leaves velocity near zero where both hypotheses make near-identical predictions.

7.2 Hyperparameter Sensitivity: Smoothing Factor γ and Inversion Threshold Λ

Table 3: Sensitivity analysis of directional EMA factor γ and structural inversion threshold Λ_{thresh} across N = 50 seeds.

Parameter Varied	Setting Tested	Mean Inversion Step	Trigger Rate	Mechanistic Assessment & Trade-off
Directional EMA Smoothing γ (at $\Lambda_{\text{thresh}} = -0.50$)	$\gamma = 0.60$	2.00 ± 0.00 steps	50/50	Vulnerable to transient sensor noise chatter
	$\gamma = 0.70$	2.00 ± 0.00 steps	50/50	Marginal noise immunity under collision spikes
	$\gamma = 0.80$ (Default)	4.00 ± 0.00 steps	50/50	Optimal balance: noise-immune & fast recovery
	$\gamma = 0.90$	7.00 ± 0.00 steps	50/50	Sluggish adaptation delay (prolongs high error)
	$\gamma = 0.95$	14.00 ± 0.00 steps	50/50	Excessive smoothing (3.5× slower polarity flip)
Inversion Threshold Λ_{thresh} (at $\gamma = 0.80$)	$\Lambda = -0.30$	2.00 ± 0.00 steps	50/50	Premature trigger risk during sharp braking
	$\Lambda = -0.40$	3.00 ± 0.00 steps	50/50	False-inversion risk during rapid decelerations
	$\Lambda = -0.50$ (Default)	4.00 ± 0.00 steps	50/50	Optimal: zero false positives on familiar physics
	$\Lambda = -0.60$	5.00 ± 0.00 steps	50/50	Delayed adaptation response
	$\Lambda = -0.70$	6.00 ± 0.00 steps	50/50	Over-conservative: delays recovery by 50%

8. CONCLUSION & OPEN REPRODUCIBILITY

DWMA v2.1 demonstrates that an unprivileged embodied agent can autonomously acquire predictive forward models, adapt to severe dynamical domain shifts, prevent catastrophic forgetting via multi-schema memory, and discriminate nonlinear physical hypotheses through epistemic active inference. Ablation experiments indicate that targeted epistemic action selection, EMA directional filtering, and discrete schema memory make measurable contributions to the evaluated outcomes, stabilizing online dynamics, accelerating hypothesis discrimination by 2.43× in median latency (2.13× on uncensored runs), and insulating prior physical models from catastrophic interference. Complete Python/JS source code, configuration files, 50-seed deterministic trajectories, and turnkey reproduction scripts (`python reproduce_all.py`) are archived in the repository for complete independent replication.

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35 foundational and contemporary references spanning world models, active inference, predictive coding, symbol grounding, and continual learning.

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